

FEN-YU HSIEH

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Education

Department of Computer Science, National Tsing Hua University (NTHU)

Hsinchu, Taiwan

B.S. IN COMPUTER SCIENCE

Sep. 2021 - June 2025

- GPA: 3.95/4.3 (3.81/4.0)

Research Interests

My research centers on **hardware–software co-design** to deliver **high throughput**, **low latency**, and **energy efficiency** via **AI accelerators**. My goal is to validate these methods on **real-world AI workloads** for **scalable**, **high-performance** deployment.

Publications

- [1] **F. Y. Hsieh***, Y. C. Teng*, D. Y. Hong, J. J. Wu, **”FPGA Co-Design for Efficient N:M Sparse and Quantized Model Inference”**, *Submitted preprint to arXiv*, 2025
- [2] S. T. Sung*, Y. W. Tang*, **F. Y. Hsieh***, R. Y. Lin*, F. Y. Hsu*, and C. T. Huang, **“HW/SW Co-Design for Efficient GPT-2 Inference on FPGA via High-Level Synthesis”**, *The 26th Workshop on Synthesis And System Integration of Mixed Information technologies (Sasimi)*, October 9-10, 2025, Nara, Japan
- [3] P. S. Wang, **F. Y. Hsieh**, J. H. Lim, M. S. Kravitz, P. C. Kuo, A. S Dagan, and N. I Shapiro, **“A Computer Vision-Based Framework for Anemia Screening Using Conjunctiva Photos”**, *Under second-round review of IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2025

Authors marked with * contributed equally as co-first authors.

Research Experience

Computer Systems Lab, Academia Sinica

Taipei, Taiwan

ADVISOR: PROF. DING-YONG HONG

- **Optimizing AI Model with Compression and Accelerator Design**

Feb. 2025 - Now

This research focuses on accelerating AI model inference and improving energy efficiency through model compression techniques and FPGA accelerator design.

1. Implemented **N:M pruning** and **low-bit quantization** to reduce model size and inference speed.
2. Adopted **hardware-friendly semi-structured sparsity** to accelerate pruning computation.
3. Stored pruned weights in **integer format** to significantly reduce model loading time and memory.
4. Implemented a **LoRA strategy** to recover and maintain model accuracy after pruning and quantization.
5. Developed an **end-to-end** benchmarking framework comparing between CPU, and GPU implementations for further improvements.

VLSI/CAD Group, NTHU

Hsinchu, Taiwan

ADVISOR: PROF. CHIH-TSUN HUANG

- **Efficient GPT-2 Inference on FPGA via High-Level Synthesis**

Jan. 2024 - Dec. 2024

This research develops hardware/software co-design techniques for accelerating GPT-2 inference on AMD Alveo U280 FPGA, focusing on high throughput and energy efficiency.

1. Implemented **GPTQ to quantize weights**, reducing model size.
2. Fine-tuned GPT-2 on **WikiText-2**, improving **perplexity** from 1.463 to 1.239.
3. Applied **High-Level Synthesis (HLS)** to reduce development time.
4. Implemented **software/hardware optimizations** to improve compute and memory efficiency.
5. Achieved a **2.16× speedup over CPU** inference while consuming only **23% of GPU power**.

ADVISOR: PROF. PO-CHIH (BRUCE) KUO

• A CV-Based Framework for Anemia Screening Using Conjunctiva Photos

Feb. 2024 - Dec. 2024

This research aims to develop a fully automated framework for hemoglobin level prediction and anemia screening using conjunctiva images, enabling reliable, non-invasive point-of-care applications.

1. Achieved a **mean absolute error of 1.57 g/dL** and **sensitivity of 0.79**, demonstrating the clinical potential of AI as a reliable, fully automated noninvasive tool for anemia screening.
2. Proposed **ROI-aware ConvNeXt** with conjunctiva masks to improve focus and reduce shortcut learning.
3. Integrated **ridge regression** with deep learning outputs to enhance prediction accuracy.
4. Used diverse conjunctiva photos with varied views and lighting to reflect real-world POC use.

Teaching Assistantship

2023 **Teaching Assistant**, 11210CS: Hardware Design and Lab., NTHU2024 **Teaching Assistant**, 11310CS: Hardware Design and Lab., NTHU2024 **Teaching Assistant**, 11310CS: Introduction to Machine Learning, NTHU**Award**

2024 **First Place (Hackers - Group Logitech) & Second Place (Grand Prize)**, Meichu Hackathon2025 **Second Place & Third Place**, Computer Science Senior Project Contest2025 **Third Place**, College of Electrical Engineering and Computer Science Senior Project Contest**Working Experience**

Embedded Systems and Chip Technology Group, ITRI

PART-TIME RESEARCH ASSISTANT

May 2024 - Nov. 2024

Computer Systems Group, Academia Sinica

RESEARCH ASSISTANT

Feb. 2025 - Present

Selected Projects

BertEncoder Accelerator Engine

May 2024 - June 2024

FINAL PROJECT OF **VLSI SYSTEM DESIGN**

- Implemented a **Verilog-based accelerator** for quantized BERT-Tiny, targeting efficient execution of multi-head attention and feedforward layers.
- Applied **SRAM-based memory management** and hardware optimizations for low-latency execution.
- Validated with RTL/gate-level simulations and Design Compiler analysis, achieving reduced power and area, demonstrating the potential of hardware acceleration for NLP tasks.

Coin Bank

Dec. 2023 - Jan. 2024

FINAL PROJECT OF **INTRODUCTION OF INTEGRATED CIRCUIT DESIGN**

- Designed a **digital coin bank** using D flip-flops, multiplexers, and finite state machines (FSMs) to store, update, and display monetary values.
- Implemented a 4-bit I/O system with binary and one-hot encoding for state transitions.
- Verified functionality through pre/post-simulation, waveform analysis, and timing checks.

Technical Skills

Programming Languages C/C++, Python, Verilog, Java, HTML, JavaScript, CSS**Hardware Design Tools** Innovus, Vivado, HSPICE, Virtuosio, SpyGlass, ModelSim, Design Compiler**Engineering Softwares** PyTorch, Git, FPGA**Operating Systems** Linux, Windows, MacOS